Department of Artificial Intelligence

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad
॥ नवोन्मेषेण देशनिर्माणम् ॥

Reinforcement Learning (AI3000)

Lecture 03

Karthik P. N.

Assistant Professor, Department of AI

Email: **pnkarthik@ai.iith.ac.in**

04 August 2026

Decision Rule

- A decision rule represents the manner in which an agent chooses an action at a given time step n
- Let $H_n = (S_0, A_0, R_1, \ldots, S_{n-1}, A_{n-1}, R_n, S_n)$ denote the history up to time n
- **Because rewards are a function of states and actions**, it is customary to take $H_n = (S_0, A_0, \ldots, S_{n-1}, A_{n-1}, S_n)$
- Let $\mathcal{H}_n = (\mathcal{S} \times \mathcal{A})^n \times \mathcal{S}$ denote the set of all feasible histories H_n
  - A **history-dependent, deterministic** decision rule at time n is a function, say π_n, such that
    $$\pi_n : \mathcal{H}_n \to \mathcal{A}, \qquad A_n = \pi_n(H_n), \qquad \mathbb{P}(\{A_n = a\} \mid H_n) = \mathbf{1}_{\{a=\pi_n(H_n)\}}.$$
  - A **history-dependent, randomised** decision rule at time n is a function, say π_n, such that
    $$\pi_n : \mathcal{H}_n \to \Delta(\mathcal{A}), \qquad \pi_n = \{\pi_n(\cdot \mid H) : H \in \mathcal{H}_n\}, \qquad \mathbb{P}(\{A_n = a\} \mid H_n) = \pi_n(a \mid H_n).$$
  - A **Markov, deterministic** decision rule at time n is a function, say π_n, such that
    $$\pi_n : \mathcal{S} \to \mathcal{A}, \qquad A_n = \pi_n(S_n), \qquad \mathbb{P}(\{A_n = a\} \mid H_n) = \mathbf{1}_{\{a=\pi_n(S_n)\}}.$$
  - A **Markov, randomised** decision rule at time n is a function, say π_n, such that
    $$\pi_n : \mathcal{S} \to \Delta(\mathcal{A}), \qquad \pi_n = \{\pi_n(\cdot \mid S) : S \in \mathcal{S}\}, \qquad \mathbb{P}(\{A_n = a\} \mid H_n) = \pi_n(a \mid S_n).$$

Policy

- A policy of the agent is simply a collection of decision rules of the form $\pi = \{\pi_n\}_{n \geq 0}$
- Depending on the nature of the component decision rules, a policy may be categorised as:
  - **History-dependent, deterministic:** π_n is history-dependent and deterministic for each n
    - We will denote the set of all history-dependent, deterministic policies by Π^{HD}
  - **History-dependent, randomised:** π_n is history-dependent and randomised for each n
    - We will denote the set of all history-dependent, randomised policies by Π^{HR}
  - **Markov, deterministic:** π_n is Markov and deterministic for each n
    - We will denote the set of all Markov, deterministic policies by Π^{MD}
  - **Markov, randomised:** π_n is Markov and randomised for each n
    - We will denote the set of all Markov, randomised policies by Π^{MR}
- A Markov policy is said to be **stationary** if it uses the same decision rule at each time step n
  - A stationary, deterministic policy: $A_n = \varphi(S_n)$ for all n, for some fixed $\varphi : \mathcal{S} \to \mathcal{A}$
  - Stationary, randomised policy: $\mathbb{P}(\{A_n = a\} \mid H_n) = \varphi(a \mid S_n)$ for all n, for some fixed $\varphi : \mathcal{S} \to \Delta(\mathcal{A})$
- Π^{HR} is the most general class of policies that we will study in the course

Ways an Agent Can Choose Actions

A policy is classified by **what information it uses** and **how it selects an action**.

INFORMATION USED \ ACTION SELECTION	**Deterministic** one prescribed action	**Randomised** sample an action
$\boldsymbol{H_n}$ **History-dependent** full history H_n	**History-dependent, deterministic** $\boldsymbol{A_n = \pi_n(H_n)}$	**History-dependent, randomised** $\boldsymbol{A_n \sim \pi_n(\cdot \mid H_n)}$
$\boldsymbol{S_n}$ **Markov** current state S_n	**Markov, deterministic** $\boldsymbol{A_n = \pi_n(S_n)}$	**Markov, randomised** $\boldsymbol{A_n \sim \pi_n(\cdot \mid S_n)}$

$$H_n = (S_0, A_0, R_1, \ldots, S_{n-1}, A_{n-1}, R_n, S_n).$$

State and State-Action Trajectories

- Given a policy $\pi = \{\pi_n\}_{n\geq 0}$, the processes

$$\{S_n\}_{n\geq 0} = \{S_0, S_1, \ldots\}, \qquad \{(S_n, A_n)\}_{n\geq 0} = \{S_0, A_0, S_1, A_1, \ldots\},$$

are respectively called the **state trajectory** and **state-action trajectory** under π

- Suppose that the PMF of S_0 is ρ, $\quad \mathbb{P}(\{S_0 = s\}) = \rho(s)$ for all s
The tuple (ρ, π) completely specifies the laws of $\{S_n\}_{n\geq 0}$ and $\{(S_n, A_n)\}_{n\geq 0}$:
  - If $\pi = \{\pi_n\}_{n\geq 0} \in \Pi^{\mathrm{HR}}$, then

$$\begin{aligned}&\mathbb{P}^{(\rho,\pi)}(\{S_0 = s_0, A_0 = a_0, \ldots, S_n = s_n, A_n = a_n\})\\ &= \rho(s_0)\cdot \pi_0(a_0 \mid s_0)\cdot P(s_1 \mid s_0, a_0)\cdot \pi_1(a_1 \mid H_1)\cdot P(s_2 \mid s_1, a_1)\cdots P(s_n|s_{n-1}, a_{n-1})\cdot \pi_n(a_n \mid H_n).\end{aligned}$$

  - If $\pi \in \Pi^{\mathrm{MR}}$, then

$$\begin{aligned}&\mathbb{P}^{(\rho,\pi)}(\{S_0 = s_0, A_0 = a_0, \ldots, S_n = s_n, A_n = a_n\})\\ &= \rho(s_0)\cdot \pi_0(a_0 \mid s_0)\cdot P(s_1 \mid s_0, a_0)\cdot \pi_1(a_1 \mid s_1)\cdot P(s_2 \mid s_1, a_1)\cdots P(s_n|s_{n-1}, a_{n-1})\cdot \pi_n(a_n \mid s_n).\end{aligned}$$

Markov Policy Induces Markov Trajectories

Theorem 1 (Markov Policy Induces Markov Trajectories)

Consider a time-homogeneous MDP with finite state and action spaces. Let $\pi = \{\pi_n\}_{n\geq 0}$ be Markov.

1. $\{(S_n, A_n)\}_{n\geq 0}$ *is a time-inhomogeneous Markov chain with TPM at time $(n+1)$ given by*
$$\mathbb{P}^{\pi}(\{S_{n+1} = s',\, A_{n+1} = a'\} \mid \{S_n = s,\, A_n = a\}) \;=\; P(s' \mid s, a) \cdot \pi_{n+1}(a' \mid s').$$
2. $\{S_n\}_{n\geq 0}$ *is a time-inhomogeneous Markov chain with TPM at time $(n+1)$ given by*
$$\mathbb{P}^{\pi}(\{S_{n+1} = s'\} \mid \{S_n = s\}) \;=\; \sum_{a\in\mathcal{A}} P(s' \mid s, a) \cdot \pi_n(a \mid s).$$
3. *If π is stationary (with $\pi_n = \varphi$ for all n), then $\{S_n\}_{n\geq 0}$ and $\{(S_n, A_n)\}_{n\geq 0}$ are time-homogeneous with*
$$\mathbb{P}^{\pi}(\{S_{n+1} = s'\} \mid \{S_n = s\}) \;=\; \sum_{a\in\mathcal{A}} P(s' \mid s, a) \cdot \varphi(a \mid s),$$
$$\mathbb{P}^{\pi}(\{S_{n+1} = s',\, A_{n+1} = a'\} \mid \{S_n = s,\, A_n = a\}) \;=\; P(s' \mid s, a) \cdot \varphi(a' \mid s').$$

Reward Process and Utility Function

Reward Process

- Given a policy $\pi = \{\pi_n\}_{n\geq 0}$, one can look at the reward process

$$\{R_n\}_{n\geq 1} \quad = \quad \{R_1, R_2, \dots\}$$

  generated under π

- The reward R_n, in general, is a stationary/time-dependent function of (S_{n-1}, A_{n-1}, S_n)
  - **Stationary, deterministic rewards:** There exists a fixed $r : \mathcal{S}\times\mathcal{A}\times\mathcal{S} \to \mathbb{R}$ such that

$$(S_{n-1}, A_{n-1}, S_n) \quad = \quad (s, a, s') \qquad \Longrightarrow \qquad R_n \quad = \quad r(s, a, s').$$

  - **Stationary, randomised rewards:** There exists a family of reward distributions $\{r_{(s,a,s')} : (s, a, s') \in \mathcal{S}\times\mathcal{A}\times\mathcal{S}\}$ such that

$$(S_{n-1}, A_{n-1}, S_n) \quad = \quad (s, a, s') \qquad \Longrightarrow \qquad R_n \quad \sim \quad r_{(s,a,s')}(\cdot).$$

  - **Time-dependent, deterministic rewards:** There exists a family of functions $\{r_n\}_{n\geq 1}$ such that

$$(S_{n-1}, A_{n-1}, S_n) \quad = \quad (s, a, s') \qquad \Longrightarrow \qquad R_n \quad = \quad r_n(s, a, s').$$

  - **Time-dependent, randomised rewards:** There exists a family of distributions $\{r^{(n)}_{(s,a,s')} : (s, a, s') \in \mathcal{S}\times\mathcal{A}\times\mathcal{S},\ \ n \geq 1\}$ such that

$$(S_{n-1}, A_{n-1}, S_n) \quad = \quad (s, a, s') \qquad \Longrightarrow \qquad R_n \quad \sim \quad r^{(n)}_{(s,a,s')}(\cdot).$$

Markov Reward Process (MRP)

- In an MDP, if $\{(S_n, A_n)\}_{n\geq 0}$ is a Markov chain (e.g., under a π that is Markov), then the reward process $\{R_n\}_{n\geq 1}$ is called a **Markov reward process**
- **Caution:** A Markov reward process DOES NOT mean that $\{R_n\}_{n\geq 1}$ is a Markov chain!
- We say that an MDP/MRP has **finite horizon** N if every policy terminates at time N
  The trajectory of states, actions, and rewards this case is as follows:

$$S_0 \xrightarrow{A_0} (S_1, R_1) \xrightarrow{A_1} (S_2, R_2) \xrightarrow{A_2} \cdots \xrightarrow{A_{N-1}} (S_N, R_N).$$

Utility Function

- A **utility function** assigns value to a reward process
- For an MDP with a finite horizon N, the utility function is defined as

$$u : \mathbb{R}^N \to \mathbb{R}, \qquad u : (R_1, \ldots, R_N) \mapsto u(R_1, \ldots, R_N).$$

- For an MDP with an infinite horizon, the utility function is defined as

$$u : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}, \qquad (R_1, R_2, \ldots) \mapsto u(R_1, R_2, \ldots).$$

- The most commonly studied utility functions are **additive**, i.e.,

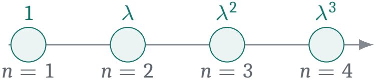
$$u(R_1, \ldots, R_N) = \sum_{n=1}^{N} u_n(R_n), \qquad \mathbb{E}[u(R_1, \ldots, R_N)] = \sum_{n=1}^{N} \mathbb{E}[u_n(R_n)],$$

$$\mathbb{E}[u(R_1, R_2, \ldots)] = \lim_{N \to \infty} \sum_{k=1}^{N} \mathbb{E}[u_n(R_n)].$$

Utility Separates Timing and Risk Preferences

$$u_n(r) = \underbrace{\lambda^{n-1}}_{\text{timing preference}} \underbrace{f(r)}_{\text{risk sensitivity}}, \qquad 0 \leq \lambda \leq 1.$$

TIMING PREFERENCE: λ



$\lambda = 1$: time-indifferent

$0 < \lambda < 1$: later rewards receive smaller weights

Smaller λ: see high rewards sooner

λ assigns weightage to timing of rewards

RISK SENSITIVITY: f

Strategy A: win 100 with probability 1.

Strategy B: win 0 with probability 0.5;
win 200 with probability 0.5.

Risk attitude	Utility	Strategy preference
Risk-averse	$f(r) = \sqrt{r}$	prefers A
Risk-neutral	$f(r) = r$	indifferent
Risk-seeking	$f(r) = r^2$	prefers B

f captures risk-sensitivity of agent